

Francisco Girbal Eiras

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Github · Google Scholar

ABOUT ME	I am a Founding AI Engineer at Magentic, building agentic AI systems for procurement and supply chain data, focused on self-improving agent harnesses and robust evaluations. Previously, I was an ML Research Scientist at Dynamo AI on compliance and safety evaluations for generative AI, and completed a PhD on Trustworthy Machine Learning at Oxford under Philip Torr, Adel Bibi and M. Pawan Kumar (Google DeepMind).	
INDUSTRY EXPERIENCE	Magentic, Founding AI Engineer	Nov. 2025 – Present
	- Introduced Magentic's first agent and lead the ongoing development of its architecture, validating design choices through experimentation against eval suites built from real procurement data. - Continuously improving system reliability through an iterative loop combining hill-climbing, feedback integration, and model routing as new data and edge cases emerge. - Embedding human-in-the-loop design throughout, balancing agent autonomy with oversight to match the risk profile of procurement and supply chain stakeholders.	
	Dynamo AI, ML Research Scientist	Nov. 2024 – Oct. 2025
	- Led the research team developing agentic red-team evaluations for LLMs, targeting custom safety specifications in conversational and tool-using agents. - Designed automated pipelines that 3x'd the coverage of red-teaming techniques within 6 months, increasing adversarial coverage and behavioral diversity across evaluation scenarios. - Contributed to core research efforts, including a first-author paper on the robustness of LLM safety judges and a medical benchmarking hallucination paper.	
	Five, Research Scientist Intern	Dec. 2022 – Jun. 2023
	Worked on efficient methods to perform zero-shot and weakly-supervised referring image segmentation (i.e., segmenting an object in an image that is referred in a natural language sentence), achieving new state-of-the-art performance in the field.	
	Five, Research Scientist Intern	Jun. 2021 – Sep. 2021
	Extended the certified robustness technique of randomized smoothing from isotropic ℓ_p balls to <i>anisotropic</i> certificates through a simplified Lipschitz analysis-based framework.	
	Five, Research Engineer	Sep. 2018 – Sep. 2020
	- Led the development of safe and scalable optimization-based motion planning algorithms, working in a team with research scientists and software engineers. - Published and presented research work developed at top tier conferences and journals within the robotics community, as well as to non-technical audiences.	
EDUCATION	University of Oxford, Oxford, UK <i>DPhil (PhD), Engineering Science, AIMS</i> Title: <i>Towards Trustworthy Machine Learning Models in Vision, Physics, and Language Applications</i> Supervisors: Prof. Philip H.S. Torr, Dr. Adel Bibi, Dr. M. Pawan Kumar (Google DeepMind)	Oct. 2020 – Oct. 2024
	University of Oxford, Oxford, UK <i>MSc, Computer Science</i> Grade: Distinction	Oct. 2017 – Sep. 2018
	Técnico Lisboa, Lisbon, Portugal <i>BSc, Electrical and Computer Engineering</i> GPA 18/20; top 2% of class	Sep. 2013 – Jul. 2016

SELECTED
PUBLICATIONS

F Eiras, E. Zemor, E. Lin, V. Mugunthan, *Know Thy Judge: On the Robustness Meta-Evaluation of LLM Safety Judges*, International Conference on Learning Representations (ICLR) Workshop Proceedings, 2025

F Eiras, A Petrov, PHS Torr, MP Kumar, A Bibi, *Do as I do (Safely): Mitigating Task-Specific Fine-tuning Risks in Large Language Models*, International Conference on Learning Representations (ICLR), 2025

F Eiras, R Bunel, K Dvijotham, A Bibi, PHS Torr, MP Kumar, *Efficient Error Certification for Physics-Informed Neural Networks*, International Conference on Machine Learning (ICML), 2024

F Eiras, A Petrov, B Vidgen, [...], T Darrell, Y Lee, J Foerster, *Near to Mid-term Risks and Opportunities of Open-Source Generative AI*, International Conference on Machine Learning (ICML, Position Paper), 2024 **[Oral]**

F Eiras, K Oksuz, A Bibi, PHS Torr, PK Dokania, *Segment, Select, Correct: a Framework for Weakly-Supervised Referring Segmentation*, European Conference on Computer Vision (ECCV) Workshop Proceedings, 2024

F Eiras, R Bunel, K Dvijotham, A Bibi, PHS Torr, MP Kumar, *Provably Correct Physics-Informed Neural Networks*, 2nd Workshop on Formal Verification of Machine Learning, International Conference on Machine Learning (ICML), 2023 **[Outstanding Paper Award]**

F Eiras, M Alfara, MP Kumar, PHS Torr, PK Dokania, B Ghanem, A Bibi, *ANCER: Anisotropic certification via sample-wise volume maximization*, Transaction of Machine Learning Research (TMLR), 2022

F Eiras, M Hawasly, SV Albrecht, S Ramamoorthy, *A Two-Stage Optimization-based Motion Planner for Safe Urban Driving*, Transaction on Robotics (T-RO), 2021

HONORS &
AWARDS

Outstanding Paper Award (by European Lighthouse on Secure and Safe AI), 2nd Workshop on Formal Verification of Machine Learning @ ICML, 2023

Honorable Mention – Using Computer Vision for Social Good, LauzHack, 2018

Graduate Research Fellowship, Institute for Systems and Robotics, 2017

Undergraduate Research Fellowship, Institute for Systems and Robotics, 2016

INVITED
TALKS

On Fine-Tuning Risks in Closed Large Language Models
OxAI (Oxford AI Society) Mini-Conference, Feb. 2024

Towards Certified Machine Learning
Columbia University, New York University, University of California, Berkeley, Stanford University, Jul. 2023

Provably Correct Physics-Informed Neural Networks
UK AI Fellows Conference (Turing Institute), May 2023

RESEARCH
INTERESTS

AI Agents Harness Optimization Agentic Evals and Hillclimbing LLM Evals and Benchmarking
Certified Machine Learning Optimization Adversarial Robustness Formal Methods

PROGRAMMING/
FRAMEWORKS

Python C/C++ PyTorch Tensorflow WandB transformers Pydantic AI Langfuse
Unslloth Qdrant Temporal git CI/CD AWS Docker React TypeScript